



CI/CD and Jenkins Hands-On Lab Guide

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Learning Objectives

By the end of this lab, participants will be able to:

- Understand CI/CD concepts and workflow
 - Explain how CI/CD pipelines work in real-world DevOps
 - Understand Jenkins and its architecture
 - Install and configure Jenkins on Ubuntu 22.04
 - Create a basic Jenkins job integrated with Git
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Learning Outcomes

After completing this lab, you will:

- Be able to design a simple CI/CD pipeline
 - Deploy Jenkins in a real environment
 - Automate code integration using Jenkins
 - Understand how Jenkins interacts with Git repositories
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Section 1: What is CI/CD?

CI/CD stands for:

- **CI (Continuous Integration)**
- **CD (Continuous Delivery / Continuous Deployment)**

♦ Continuous Integration (CI)

Developers frequently merge code into a shared repository. Each integration is verified using automated builds and tests.

👉 Goal: Detect issues early

◆ Continuous Delivery (CD)

Code is automatically prepared for release to production.

◆ Continuous Deployment

Every change that passes testing is automatically deployed.

How CI/CD Works (Step-by-Step)

1. Developer writes code
2. Pushes code to Git repository (GitHub/GitLab)
3. CI/CD tool (Jenkins) detects changes
4. Code is pulled into build server
5. Build process starts
6. Automated testing runs
7. Artifacts are created
8. Deployment happens (optional)

👉 This entire process is called a **Pipeline**

Section 2: What is Jenkins?

Jenkins is an open-source automation server used to implement CI/CD pipelines.

◆ Key Features:

- Open source
 - Plugin-based architecture
 - Easy integration with Git, Docker, Kubernetes
 - Supports pipelines (CI/CD automation)
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Jenkins Architecture

Jenkins follows a **Master-Agent Architecture**

◆ Components:

1. Jenkins Master

- Controls everything
- Schedules jobs
- Manages plugins

2. Jenkins Agent (Worker Nodes)

- Executes jobs
- Can be multiple

3. Jenkins Job / Pipeline

- Task or automation process

4. Plugins

- Extend Jenkins functionality



Section 3: Hands-On Lab – Install Jenkins on Ubuntu 22.04



Prerequisites

- Ubuntu 22.04 Server
- Minimum 2GB RAM
- Sudo privileges
- Internet connectivity



Step 1: Update System

```
sudo apt update -y
```



Step 2: Install Java (Jenkins Requirement)

```
sudo apt install fontconfig openjdk-21-jre
```

Verify:

```
java -version
```

Step 3: Add Jenkins and Install Jenkins

```
sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc  
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key  
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc]  
https://pkg.jenkins.io/debian-stable binary/ | sudo tee  
/etc/apt/sources.list.d/jenkins.list > /dev/null  
sudo apt update  
sudo apt install jenkins
```

Step 4: Start Jenkins Service

```
sudo systemctl start jenkins  
sudo systemctl enable jenkins
```

Check status:

```
sudo systemctl status jenkins
```

Step 5: Open Port 8080 (Firewall)

```
sudo ufw allow 8080
```

Step 6: Access Jenkins

Open browser:

```
http://<your-server-ip>:8080
```

Step 7: Unlock Jenkins

Get initial password:

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

Paste it in browser

Step 8: Install Suggested Plugins

Click: **Install Suggested Plugins**

Step 9: Create Admin User

Fill:

- Username
 - Password
 - Email
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Section 4: Create First Jenkins Job

◆ Steps:

1. Go to Dashboard
 2. Click **New Item**
 3. Enter name → Select **Freestyle Project**
 4. Source Code Management → Select **Git**
 5. Add GitHub repository URL
 6. Click **Build Now**
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Verification

- Check Console Output
 - Verify code is cloned in workspace
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Conclusion

You have successfully:

- Understood CI/CD concepts
 - Installed Jenkins on Ubuntu
 - Created your first CI job
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